WaterLink

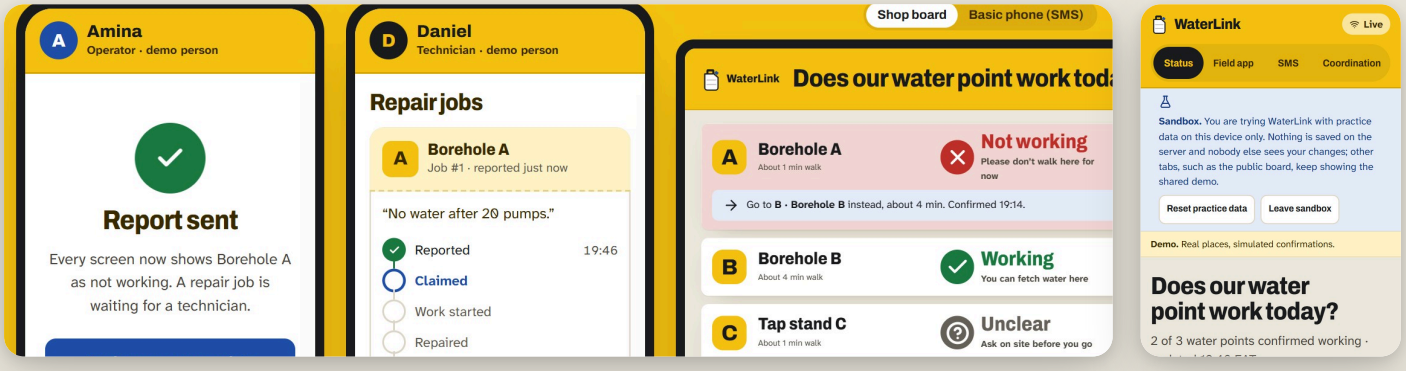
Does our water point work today?

WaterLink shows the **last confirmed state** of every water point before people walk, and makes every outage a **job with a named person** that only counts as repaired once the pump was **tested on site**.

The problem is real and current

- Up to **one in three hand pumps** does not function at any one time across Africa; people can be without clean water for at least 2–3 days (UNICEF Office of Innovation, “Smart Pumps”).
- **2.1 billion people** still lack safely managed drinking water; 1 in 4 worldwide (WHO/UNICEF JMP report, 25 Aug 2025).
- The public data cannot say what works today: of **39,959** Kenyan survey records the newest is from 30 Nov 2024, the median **5.7 years** old (WPdx, our calculation).

The prototype: working today, not a mock-up



- Public board and household view, English and Kiswahili (draft), printable.
- Report → job → blockers → repair → **test on site**, with history.
- Stale, conflicting or offline states never look working.
- Alternatives only with a checked route; add a water point by tap on the map.
- SMS simulation, replay of real 2014 pump data, practice mode without login.
- 85 unit and 37 browser tests on every push.

Numbers behind the assumptions

- Hand pump failures: up to 1 in 3 (UNICEF).
- Kyuso, Kenya, top complaints: repair cost 19%, repair time 17%, breakdowns 17%.
- Mobile phones: 48.6% of rural, 64.6% of urban people (Housing Survey 2023/24).
- Kenya, last survey: 21% non-functional, 27% in Busia (WPdx).
- Downtime with sensors and a repair service: 27 days to under 3 (WASREB 2014).
- **Open, no evidence yet:** will operators confirm often enough, will anyone pay. No pilot, no impact claim.

IdeaLAB: we would potentially agree to take part if we are selected.

Try it, no login: waterlink-demo.waterlink.workers.dev/?view=stage&sandbox=1
Code: github.com/OsmanTuerkmen/WaterLink
Team: Recep Baş (basreci@gmail.com), Osman Türkmen · Zero/One Hackathon, WU Vienna, 22 Oct 2026, Social Impact

People, confirmations and routes in the demo are simulated; places are from WPdx surveys. “Working” never means safe to drink.